

Balanced Armature Drivers

These are the smallest type of drivers and are preferred in canalphones and IEMs (In Ear Monitors). These drivers fit into the ear canals and reduce the amount of ambient noise which get into the ears, thereby producing some form passive noise cancellation.

Planar Drivers

Planar drivers or ortho-dynamic drivers work by moving a membrane fixed between two powerful magnets which induce a coil fixed on the membrane. Planar drivers usually need more power to output the same volume as dynamic drivers. These are used in high end headphones, which will often need external amplifiers to deliver true sound.

Electrostatic Drivers

Electrostatic drivers are a relatively new technology which use a thin electrically charged diaphragm to produce sound. These are very expensive and need special amplifiers to work, but deliver the truest sound. The amplifier needed is called an energizer, which delivers charge to the membrane and the plates present on either side of the membrane. As the charges change, the membrane vibrates to reflect the changes.

To wrap things up

With headphones, you get what you pay for. The more you pay, the audio manufacturers are able to add better drivers, DACs and more electronics, which are able to recreate the studio sound accurately. Headphones are designed broadly for specific purposes and you can often find one that suits your need, but it is difficult to find one which can suit all needs. A true all-rounder is still a myth. Headphones are evolving at a rapid pace, with people preferring more personal sound and more portability. Bluetooth headphones are rapidly rising in the smartphone market, with multiple smartphone manufacturers trying out courageous moves by promising better sound. While audiophiles nit-pick at small and sometimes insignificant details, they are in constant pursuit of the perfect sound.

Headphones are often paired with DACs/AMPs which enhance the sound by providing more power to the drivers, power which cannot be provided by the regular audio sources like smartphones, tablets and PCs. Headphones also depend heavily on the audio source. If you buy really expensive headphones, but pair them with devices which have cheap